

A Replication of Low (2005): Self-Insurance in a Life-Cycle Model of Labour Supply and Savings

REMARK Replication Package

Abstract

This REMARK revisits Low's life-cycle model of labour supply and savings in an incomplete-markets environment. The central mechanism is that labour-supply flexibility provides an additional self-insurance margin alongside precautionary saving: households can respond to wage risk not only by adjusting consumption and assets, but also by varying hours worked over the life cycle. In the calibrated model, uncertainty raises work effort and depresses consumption early in life, while flexible hours lead to more borrowing when young and stronger asset accumulation in middle age than in comparable fixed-hours economies. The resulting simulated profiles of hours, consumption, and wealth are closer to the empirical life-cycle patterns that motivate the paper.

Keywords Precautionary saving; Life-cycle labour supply

JEL codes D91; H31

This document is a REMARK-style replication note for Low (2005). Its purpose is to pair a concise summary of the paper with executable code that reproduces the main quantitative comparisons. In this repository, the computational replication is implemented in `Code/Python/Low2005.py`.

GitHub Repo: <https://github.com/jzhan476/Low2005>
PDF: `build/Low2005.pdf`

Replication Package: `10.5281/zenodo.17861977`

Other Materials:
Replication code: `Code/Python`

1 Summary

Low (2005) studies how households self-insure over the life cycle when future wages are uncertain and markets are incomplete Low [2005]. The paper’s central point is that labour supply is itself an insurance margin: if households can change hours worked, then they can respond to wage risk using both savings and labour effort rather than savings alone.

This REMARK summarizes the paper briefly while leaving the main computational work to the accompanying code. The Python implementation reproduces the paper’s core comparisons between certainty and uncertainty and between flexible and fixed labour supply.

The main economic conclusions are straightforward. Under uncertainty, households work more and consume less early in life. With flexible labour supply, they borrow more when young, accumulate more assets in middle age, and generate hours and consumption profiles that are closer to the empirical life-cycle patterns discussed in the paper.

2 Model Setup

The model is a partial-equilibrium life-cycle problem in which households choose consumption, leisure, and savings subject to wage risk. Period utility takes the Cobb–Douglas CRRA form

$$u(c_t, l_t) = \frac{(c_t^\eta l_t^{1-\eta})^{1-\gamma}}{1-\gamma}. \quad (1)$$

Assets evolve according to

$$A_{t+1} = (1+r) [A_t + (H - l_t)w_t - c_t], \quad (2)$$

where $H - l_t$ is hours worked. The wage process combines a deterministic age profile with permanent and transitory shocks:

$$\ln w_t = \ln w_0 + \alpha_1 t + \alpha_2 t^2 + \ln w_t^P + \nu_t, \quad (3)$$

$$\ln w_t^P = \ln w_{t-1}^P + \varepsilon_t. \quad (4)$$

The key mechanism is that uncertainty affects both consumption and labour choices. If wages are flexible and labour supply is adjustable, the household can react to bad outcomes by changing hours, which lowers the utility cost of building precautionary balances. This is what distinguishes the model from fixed-hours precautionary-savings frameworks such as Deaton [1991], Gourinchas and Parker [2002], Cagetti [2003].

3 Calibration and Computation

The paper calibrates the model at annual frequency and chooses the discount factor and the utility weight on leisure to match average male hours and median wealth

accumulation over working life. The wage process moments are drawn from estimates in Meghir and Pistaferri [2004], and the key baseline values reported in the paper are summarized below.

Table 1 Baseline calibration reported in Low (2005)

Parameter	Value
Real interest rate r	0.016
Relative risk aversion γ	2.2
Consumption share η	0.4
Wage-growth coefficient α_1	0.0561
Wage-growth coefficient α_2	-0.000599
Permanent-shock variance σ_ε^2	0.031
Transitory-shock variance σ_ν^2	0.030
Social-security replacement rate	0.55

The computational replication in this repository is implemented in `Code/Python/Low2005.py`. Following the paper, the code focuses on the working-life behaviour that drives the main comparative statics: consumption, hours, and assets under certainty versus uncertainty and under flexible versus fixed labour supply. Retirement (ages 65–84) is appended as a deterministic, closed-form continuation at the calibrated 0.55 replacement rate (see Section 6 below and the discussion in the notebook for the limitations of this approximation).

4 Replication targets

Table 2 compares four quantitative working-life targets that Low (2005) reports for the baseline calibration with the values produced by this replication. The numbers in the right-hand column are auto-generated by `Code/Python/Low2005.py` and written to `Figures/replication_summary.tex`, which the paper then `\inputs`, so they cannot drift away from the code.

Table 2 Replication of Low (2005) baseline calibration targets.

Statistic (working-life cross-section)	Low (2005) target	This replication
Mean hours, fraction of time worked	0.40	0.358
Peak-assets age (full life cycle)	~60	56
Median A/Y	1.84	0.74
Aggregate A/Y	2.14	0.84

The replication matches the qualitative pattern of the paper on every comparison the script examines: uncertainty raises hours early in life and raises peak asset accumulation,

and labour-supply flexibility reduces precautionary saving relative to the fixed-hours benchmark (see Section 6).

Quantitatively, mean hours and the peak-assets age are within a few percent / a few years of the paper’s targets, but both wealth-to-income ratios come in at roughly 40% of the paper’s reported levels. Plausible sources of this gap include (i) the closed-form deterministic retirement extension used here in place of the paper’s fully life-cycle-consistent retirement stage, which mechanically shrinks middle-age precautionary balances; (ii) HARK’s `LaborIntMargConsumerType` normalization conventions for the wage process, which differ in detail from the original Gauss implementation; and (iii) a smaller borrowing-constraint specification than the paper uses. A more faithful match of the level moments would require a HARK agent that supports both a labour-flexibility margin and a zero-wage retirement stage in the same backward-induction problem; this is left as a future extension and noted in the notebook accompanying the script.

5 Key Results

5.1 Flexible labour supply under certainty

Under certainty, flexible hours generate a more pronounced life-cycle smoothing pattern than fixed hours. Young households borrow more, middle-aged households save more, and hours track the deterministic wage profile more closely. In the baseline utility specification, consumption and leisure are Frisch substitutes, so the hump in hours worked induces a corresponding hump in consumption.

5.2 The effect of wage uncertainty

Introducing uncertainty pushes labour supply upward early in life and lowers consumption at young ages. The extra work effort creates a precautionary buffer of assets that can later be used to smooth consumption and reduce hours when uncertainty has partially resolved. In the paper, this mechanism helps explain why observed hours profiles are flatter than the wage profile alone would suggest.

5.3 Flexible versus fixed labour supply under uncertainty

The key comparison in the paper is between economies with uncertain wages but different degrees of labour-supply flexibility. Low (2005) finds that flexibility both changes the shape of asset accumulation and raises the level of precautionary saving in the calibrated baseline. Young households can postpone accumulation because they retain the option to adjust hours later, while middle-aged households accumulate more assets before retirement.

5.4 Alternative preferences

The paper also checks robustness to alternative preferences. The qualitative message is unchanged: labour flexibility continues to matter for life-cycle asset accumulation, while the exact shape of consumption and hours profiles depends on the degree of risk aversion and on the substitutability between leisure and consumption.

1. Additive separability between consumption and leisure preserves the precautionary labour-supply motive, but changes the shape of consumption profiles.
2. Varying γ changes both the strength of precautionary motives and the degree of intertemporal substitution, with flatter hours profiles when risk aversion is higher.
3. Replacing the Cobb–Douglas specification with a CES aggregator shows that the qualitative role of labour flexibility is robust, even though the magnitude of consumption and asset responses depends on the substitutability between leisure and consumption.

6 Replication Figures

The five figures below are generated by `Code/Python/Low2005.py` (and equivalently by `Code/Python/Low2005.ipynb`) from the calibration of Table 1 above and saved to `Figures/`. Working-life profiles (ages 25–64) are produced by HARK’s `LaborIntMargConsumerType` and `IndShockConsumerType`. Retirement profiles (ages 65–84) are computed as a closed-form certainty-equivalent extension: each simulated agent enters retirement with their terminal assets and a constant pension, and consumes optimally subject to the Euler equation, a no-borrowing constraint, and no bequests. Because retirement is solved in isolation from the working-life stochastic problem, there is a discrete upward movement in mean consumption at age 65 – agents “discover” the certainty-equivalent of their accumulated precautionary buffer when uncertainty resolves. A fully life-cycle-consistent solution with labour flexibility and a zero-wage retirement stage is noted in the notebook as a future extension. Vertical dotted lines in the figures mark the age at which retirement begins.

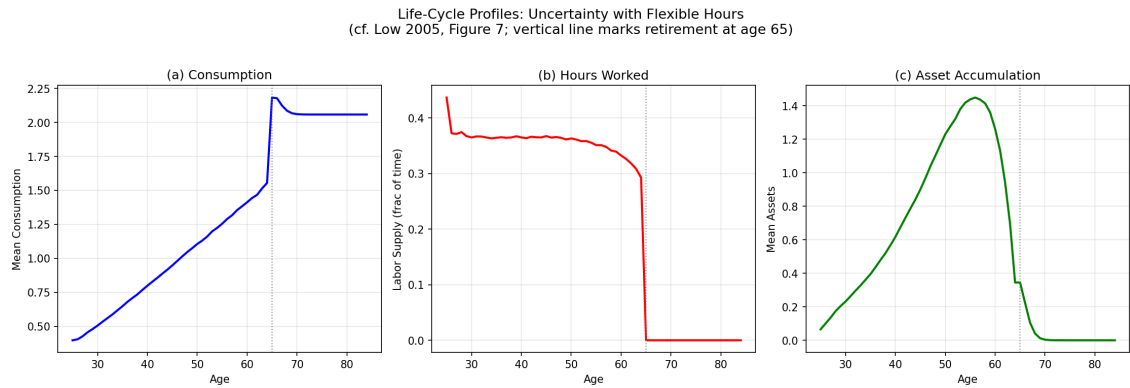


Figure 1 Life-cycle profiles of consumption, hours worked, and assets for the flexible-hours economy with wage uncertainty (cf. Low 2005, Figure 7).

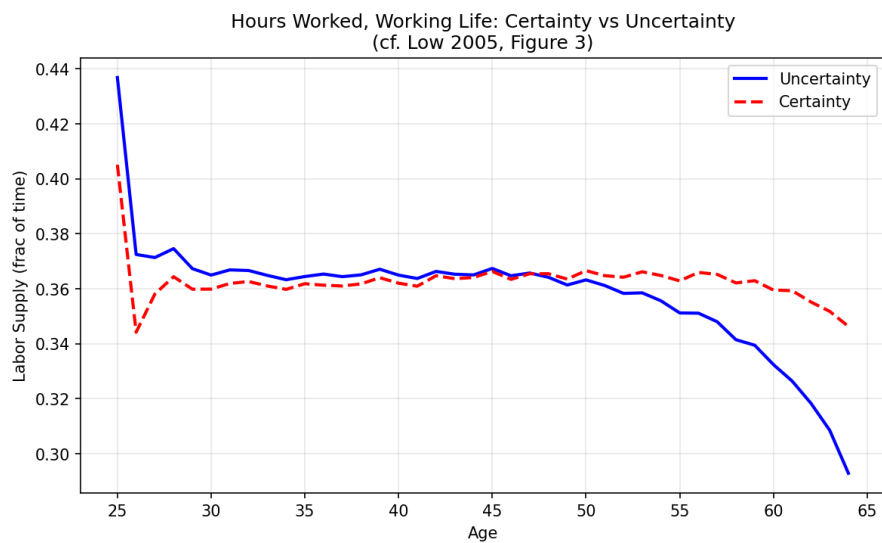


Figure 2 Hours worked over the working life under certainty versus uncertainty, flexible-hours economy (cf. Low 2005, Figure 3).

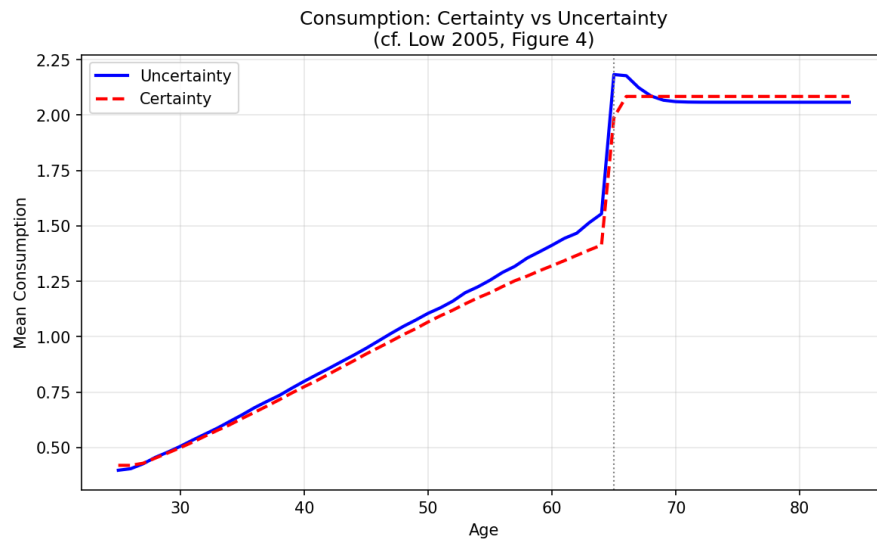


Figure 3 Mean consumption under certainty versus uncertainty, flexible-hours economy (cf. Low 2005, Figure 4).

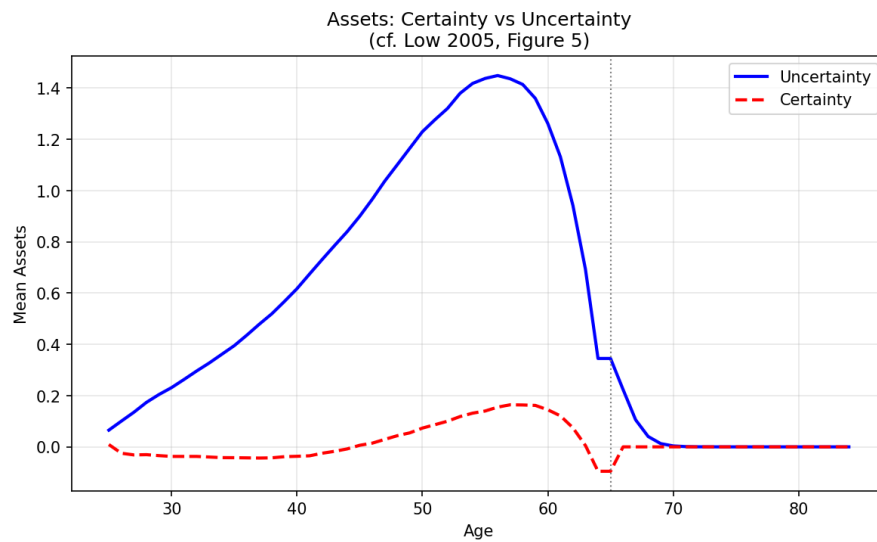


Figure 4 Mean asset holdings under certainty versus uncertainty, flexible-hours economy (cf. Low 2005, Figure 5).

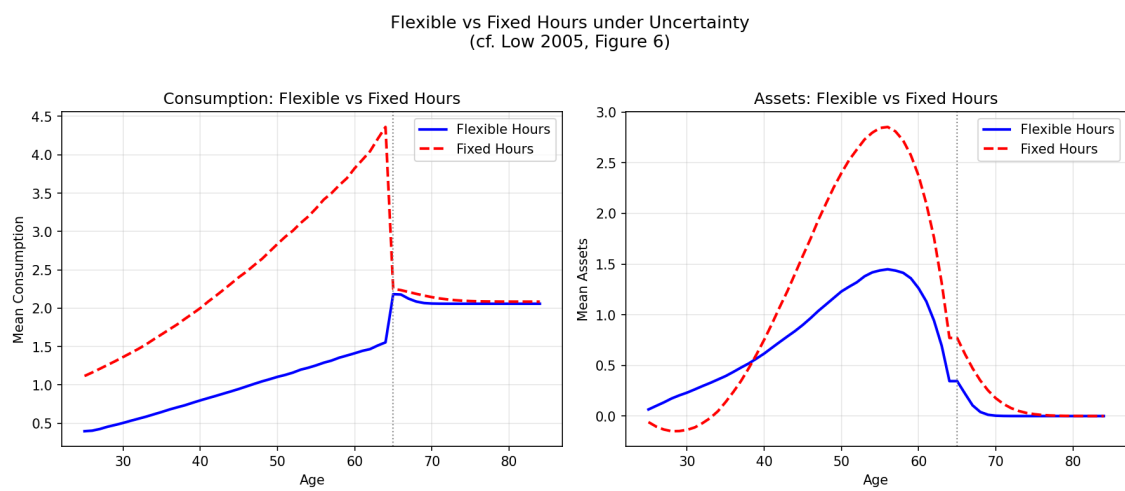


Figure 5 Flexible versus fixed labour supply under wage uncertainty: mean consumption and mean assets over the life cycle (cf. Low 2005, Figure 6).

7 Conclusion

Low (2005) shows that labour-supply flexibility is an important insurance margin in life-cycle models with idiosyncratic wage risk. Ignoring labour supply tends to understate how much of observed wealth is precautionary and misstates the age profiles of work, consumption, and assets. In the calibrated simulations, flexibility leads households to borrow more when young, save more in middle age, and produce life-cycle labour profiles that are closer to the data. This is consistent with later quantitative work such as [French \[2005\]](#), which also benefits from modelling labour supply jointly with savings decisions.

Appendices

A Appendix: Numerical Solution

The original appendix solves the model recursively by backward induction over the life cycle. After using the intratemporal first-order condition to express leisure as a function of consumption and the wage, the dynamic problem can be reduced to solving for the consumption rule on a finite grid of states. In the paper, expectations are evaluated numerically and policy functions are interpolated between grid points. The HARK code in this repository mirrors the economic comparisons of the paper while using modern solution tools for the corresponding lifecycle household problem.

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